

SIMATS ENGINEERING

ASSESSMENT TOOL 1

COURSE NAME: COMPUTER NETWORKS
COURSE CODE: CSA0716

COURSE OUTCOME COVERED

CO2: Configure IP addressing schemes, subnetting, and basic routing techniques using simulation tool, for efficient network design and topology. (BL3)

Assessment Tool Weightage: 40%

Annexure A

ANALYTICAL PROBLEM SOLVING

1. A university network is assigned the IP address block 192.168.20.0/24 for its internal communication. The network administrator has decided to divide this network into subnets for three different labs using fixed subnet masks. Lab A is assigned a /26 subnet, Lab B is assigned a /27 subnet, and Lab C is assigned a /28 subnet. Based on the given subnet masks, apply subnetting concepts to determine the network address of each lab. Further, identify the valid range of host IP addresses and the broadcast address for each subnet. Finally, calculate the total number of usable host IP addresses available in each lab's subnet.

1. Problem Understanding

A university has been assigned the IPv4 network 192.168.20.0/24. The network administrator must divide this network into separate subnets for Lab A, Lab B, and Lab C using the given subnet masks. The objective is to determine the network address, broadcast address, valid host IP range, and usable host addresses for each subnet using subnetting concepts.

2. Given Data

Parameter	Value
Original Network	192.168.20.0/24
Lab A Prefix	/26
Lab B Prefix	/27
Lab C Prefix	/28

3. Method Selection

The subnetting process is carried out using the given subnet masks. Subnets are allocated sequentially from the available 192.168.20.0/24 network, beginning with the largest subnet and continuing with the remaining address space. This ensures efficient IP address utilization while preventing address overlap.

4. Formula Used

Host Bits

Host Bits = 32 – Prefix Length

Total Addresses

Total Addresses = $2^{(\text{Host Bits})}$

Usable Host Addresses

Usable Hosts = $2^{(\text{Host Bits})} - 2$

Block Size

Block Size = 256 – Last Octet of Subnet Mask

5. Step-by-Step Solution

A) Lab A

Prefix Length

/26

Subnet Mask

255.255.255.192

Host Bits

$$32 - 26 = 6$$

Block Size

$$256 - 192 = 64$$

Total Addresses

$$2^6 = 64$$

Usable Hosts

$$64 - 2 = 62$$

Network Address

192.168.20.0

Valid Host IP Range

192.168.20.1 – 192.168.20.62

Broadcast Address

192.168.20.63

B) Lab B

Prefix Length

/27

Subnet Mask

255.255.255.224

Host Bits

$$32 - 27 = 5$$

Block Size

$$256 - 224 = 32$$

Total Addresses

$$2^5 = 32$$

Usable Hosts

$$32 - 2 = 30$$

Next Available Network

After Lab A ends at 192.168.20.63, the next subnet starts at:

192.168.20.64

Network Address

192.168.20.64

Valid Host IP Range

192.168.20.65 – 192.168.20.94

Broadcast Address

192.168.20.95

C) Lab C

Prefix Length

/28

Subnet Mask

255.255.255.240

Host Bits

$32 - 28 = 4$

Block Size

$256 - 240 = 16$

Total Addresses

$2^4 = 16$

Usable Hosts

$16 - 2 = 14$

Next Available Network

After Lab B ends at 192.168.20.95, the next subnet starts at:

192.168.20.96

Network Address

192.168.20.96

Valid Host IP Range

192.168.20.97 – 192.168.20.110

Broadcast Address

192.168.20.111

6. Engineering Justification

The subnet allocation is performed from the largest subnet to the smallest subnet to optimize IP address utilization and avoid address fragmentation. Each laboratory receives an independent subnet, which reduces broadcast traffic, improves network security, and simplifies network administration. The remaining address space is preserved for future expansion.

7. Final Result

Laboratory	Prefix	Subnet Mask	Network Address	Valid Host IP Range	Broadcast Address	Usable Hosts
Lab A	/26	255.255.255.192	192.168.20.0	192.168.20.1 – 192.168.20.62	192.168.20.63	62
Lab B	/27	255.255.255.224	192.168.20.64	192.168.20.65 – 192.168.20.94	192.168.20.95	30
Lab C	/28	255.255.255.240	192.168.20.96	192.168.20.97 – 192.168.20.110	192.168.20.111	14

8. Engineering Interpretation

The network 192.168.20.0/24 has been successfully subnetted into three independent subnets according to the specified requirements. Lab A supports 62 usable hosts, Lab B supports 30 usable hosts, and Lab C supports 14 usable hosts. The subnet allocation is efficient, minimizes address wastage, reduces broadcast traffic, and provides better network organization and scalability for future expansion.

2. A network administrator has configured two systems with the following details: System 1 has IP address 192.168.1.130 with subnet mask 255.255.255.192, and System 2 has IP address 192.168.1.190 with subnet mask 255.255.255.192. Using the given configuration, apply subnetting techniques to determine the subnet range, network address, and broadcast address for each system. Identify the valid host IP range for both subnets based on the given subnet mask. Calculate whether each IP address falls within its respective subnet range. Also, determine the number of usable hosts addresses available in each subnet.

1. Problem Understanding

Two computer systems are configured with the IP addresses **192.168.1.130** and **192.168.1.190**, both using the subnet mask **255.255.255.192 (/26)**. The objective is to determine the subnet range, network address, broadcast address, valid host range, verify whether each IP belongs to its subnet, and calculate the number of usable host addresses.

2. Given Data

Parameter	System 1	System 2
IP Address	192.168.1.130	192.168.1.190
Subnet Mask	255.255.255.192	255.255.255.192
Prefix Length	/26	/26

3. Method Selection

The subnet is determined by applying the **/26 subnet mask** to the given IP addresses. The block size is calculated first, followed by identifying the subnet in which each IP address falls. The network address, broadcast address, valid host range, and usable host addresses are then determined.

4. Formula Used

Host Bits

Host Bits = 32 – Prefix Length

Block Size

Block Size = 256 – Last Octet of Subnet Mask

Total Addresses

$2^{(\text{Host Bits})}$

Usable Hosts

$2^{(\text{Host Bits})} - 2$

5. Step-by-Step Solution

Step 1: Determine Block Size

Subnet Mask

255.255.255.192

Last Octet = **192**

Block Size

= $256 - 192$

= **64**

Therefore, the subnet ranges are:

Subnet Number	Address Range
Subnet 1	192.168.1.0 – 192.168.1.63
Subnet 2	192.168.1.64 – 192.168.1.127
Subnet 3	192.168.1.128 – 192.168.1.191
Subnet 4	192.168.1.192 – 192.168.1.255

Step 2: Analyze System 1

IP Address

192.168.1.130

Since **130** lies between **128 and 191**, it belongs to:

Subnet 192.168.1.128/26

Network Address

192.168.1.128

Broadcast Address

192.168.1.191

Valid Host Range

192.168.1.129 – 192.168.1.190

Verification

The IP address **192.168.1.130** lies within the valid host range.

Therefore, System 1 belongs to this subnet.

Step 3: Analyze System 2

IP Address

192.168.1.190

Since **190** lies between **128 and 191**, it also belongs to:

Subnet 192.168.1.128/26

Network Address

192.168.1.128

Broadcast Address

192.168.1.191

Valid Host Range

192.168.1.129 – 192.168.1.190

Verification

The IP address **192.168.1.190** is the **last valid host address** before the broadcast address.

Therefore, System 2 also belongs to this subnet.

Step 4: Calculate Usable Host Addresses

Host Bits

$= 32 - 26$

$= 6$

Total Addresses

$$= 2^6$$

$$= 64$$

Usable Hosts

$$= 64 - 2$$

$$= 62$$

6. Engineering Justification

Both systems are configured with the same subnet mask (/26) and fall within the subnet **192.168.1.128/26**. Since both IP addresses belong to the same network, they can communicate directly without requiring routing through another network. This subnetting approach reduces broadcast traffic while ensuring efficient IP address utilization.

7. Final Result

Parameter	System 1	System 2
IP Address	192.168.1.130	192.168.1.190
Prefix	/26	/26
Subnet Mask	255.255.255.192	255.255.255.192
Network Address	192.168.1.128	192.168.1.128
Broadcast Address	192.168.1.191	192.168.1.191
Valid Host Range	192.168.1.129 – 192.168.1.190	192.168.1.129 – 192.168.1.190
IP Belongs to Subnet	Yes	Yes
Usable Host Addresses	62	62

8. Engineering Interpretation

Both systems are located within the subnet **192.168.1.128/26** and share the same subnet mask. As a result, they can communicate directly without passing through a router. The subnet provides **62 usable host addresses**, making it suitable for small or medium-sized departmental networks. Proper subnetting improves network efficiency, reduces broadcast traffic, and simplifies network management.

9. Calculation Summary

Parameter	Value
Prefix Length	/26
Subnet Mask	255.255.255.192
Host Bits	6
Block Size	64
Total Addresses	64
Usable Host Addresses	62

3. An organization has been allocated the IP block **172.16.0.0/16** for its internal network. The network administrator has already decided the subnet masks for each department as follows: HR → /25, Sales → /26, IT → /27, and Admin → /28. Using the given subnet masks, apply subnetting techniques to assign suitable subnet addresses for each department. Determine the network address, valid host IP range, and broadcast address for each subnet. Based on the allocation, calculate the number of usable host addresses in each department. Finally, identify the remaining unused IP address range within the given network block.

1. Problem Understanding

An organization has been allocated the IPv4 network **172.16.0.0/16** for its internal communication. Different departments require different subnet sizes based on the assigned prefix lengths. The objective is to allocate suitable subnet addresses, determine the network address, valid host range, broadcast address, calculate the usable host addresses, and identify the remaining unused IP address range within the given network block.

2. Given Data

Parameter	Value
Original Network	172.16.0.0/16
HR Department	/25
Sales Department	/26
IT Department	/27
Admin Department	/28

3. Method Selection

The subnet allocation is performed sequentially from the available **172.16.0.0/16** network. The larger subnet is assigned first, followed by progressively smaller subnets. This method ensures efficient utilization of the available IP address space while preventing overlap between departments.

4. Formula Used

Host Bits

Host Bits = $32 - \text{Prefix Length}$

Total Addresses

Total Addresses = $2^{(\text{Host Bits})}$

Usable Host Addresses

Usable Hosts = $2^{(\text{Host Bits})} - 2$

Block Size

Block Size = $256 - \text{Last Octet of Subnet Mask}$

5. Step-by-Step Solution

A) HR Department

Prefix Length

/25

Subnet Mask

255.255.255.128

Host Bits

$$32 - 25 = 7$$

Block Size

$$256 - 128 = 128$$

Total Addresses

$$2^7 = 128$$

Usable Hosts

$$128 - 2 = 126$$

Network Address

172.16.0.0

Valid Host Range

172.16.0.1 – 172.16.0.126

Broadcast Address

172.16.0.127

B) Sales Department

Prefix Length

/26

Subnet Mask

255.255.255.192

Host Bits

$$32 - 26 = 6$$

Block Size

$$256 - 192 = 64$$

Total Addresses

$$2^6 = 64$$

Usable Hosts

$$64 - 2 = 62$$

Next Available Network

172.16.0.128

Network Address

172.16.0.128

Valid Host Range

172.16.0.129 – 172.16.0.190

Broadcast Address

172.16.0.191

C) IT Department

Prefix Length

/27

Subnet Mask

255.255.255.224

Host Bits

$$32 - 27 = 5$$

Block Size

$$256 - 224 = 32$$

Total Addresses

$$2^5 = 32$$

Usable Hosts

$$32 - 2 = 30$$

Next Available Network

172.16.0.192

Network Address

172.16.0.192

Valid Host Range

172.16.0.193 – 172.16.0.222

Broadcast Address

172.16.0.223

D) Admin Department

Prefix Length

/28

Subnet Mask

255.255.255.240

Host Bits

$$32 - 28 = 4$$

Block Size

$$256 - 240 = 16$$

Total Addresses

$$2^4 = 16$$

Usable Hosts

$$16 - 2 = 14$$

Next Available Network

172.16.0.224

Network Address

172.16.0.224

Valid Host Range

172.16.0.225 – 172.16.0.238

Broadcast Address

172.16.0.239

6. Remaining Unused IP Address Range

After allocating the Admin subnet, the remaining IP addresses are available from:

Starting Address

172.16.0.240

Ending Address

172.16.255.255

This address space can be reserved for future departments or additional subnet allocations.

7. Engineering Justification

The subnet allocation follows the principle of assigning larger subnets before smaller ones, ensuring efficient IP address utilization and avoiding address fragmentation. Each department is placed in an independent subnet, reducing broadcast traffic and improving network performance, security, and scalability.

8. Final Result

Department	Prefix	Subnet Mask	Network Address	Valid Host Range	Broadcast Address	Usable Hosts
HR	/25	255.255.255.128	172.16.0.0	172.16.0.1 – 172.16.0.126	172.16.0.127	126
Sales	/26	255.255.255.192	172.16.0.128	172.16.0.129 – 172.16.0.190	172.16.0.191	62
IT	/27	255.255.255.224	172.16.0.192	172.16.0.193 – 172.16.0.222	172.16.0.223	30
Admin	/28	255.255.255.240	172.16.0.224	172.16.0.225 – 172.16.0.238	172.16.0.239	14

Remaining Unused Address Space

Starting Address	Ending Address
172.16.0.240	172.16.255.255

9. Calculation Summary

Department	Host Bits	Block Size	Total Addresses	Usable Hosts
HR	7	128	128	126
Sales	6	64	64	62
IT	5	32	32	30
Admin	4	16	16	14

10. Engineering Interpretation

The organization has successfully subnetted the **172.16.0.0/16** network into four independent departmental networks based on the required subnet masks. HR receives **126 usable host addresses**, Sales receives **62**, IT receives **30**, and Admin receives **14**. The remaining address range from **172.16.0.240 to 172.16.255.255** remains available for future expansion. This subnetting strategy improves network organization, minimizes broadcast traffic, enhances security through network segmentation, and supports future scalability.

4. A company uses the IP address **172.16.10.0/24** and has implemented subnetting using a **/27 subnet mask** for its internal departments. Using the given subnet mask, apply subnetting techniques to calculate the total number of subnets created and the number of usable hosts per subnet. Determine the subnet to which the IP address **172.16.10.78** belongs. Identify the network address and broadcast address for that subnet. Also, determine whether the IP address **172.16.10.95** falls within the same subnet range based on the given configuration.

1. Problem Understanding

A company has been assigned the IPv4 network **172.16.10.0/24** and has implemented subnetting using a **/27 subnet mask**. The objective is to determine the total number of subnets, usable hosts per subnet, identify the subnet containing the IP address **172.16.10.78**, determine its network and broadcast addresses, and verify whether **172.16.10.95** belongs to the same subnet.

2. Given Data

Parameter	Value
Original Network	172.16.10.0/24
Subnet Mask	255.255.255.224
Prefix Length	/27
Given IP Address	172.16.10.78
Second IP Address	172.16.10.95

3. Method Selection

The subnet is determined using the given **/27 subnet mask**. First, the block size is calculated to identify all subnet ranges. The given IP addresses are then mapped to their respective subnet, and the network address, broadcast address, and host range are determined.

4. Formula Used

Number of Borrowed Bits

$$\text{Borrowed Bits} = 27 - 24 = 3$$

Total Number of Subnets

$2^{(\text{Borrowed Bits})}$

Host Bits

$32 - \text{Prefix Length}$

Total Addresses per Subnet

$2^{(\text{Host Bits})}$

Usable Host Addresses

$2^{(\text{Host Bits})} - 2$

Block Size

$256 - \text{Last Octet of Subnet Mask}$

5. Step-by-Step Solution

Step 1: Calculate the Total Number of Subnets

Borrowed Bits

$= 27 - 24$

$= 3$

Total Number of Subnets

$= 2^3$

$= 8 \text{ Subnets}$

Step 2: Calculate the Number of Usable Hosts

Host Bits

$= 32 - 27$

$= 5$

Total Addresses

$= 2^5$

$= 32$

Usable Host Addresses

$$= 32 - 2$$

= **30 Hosts**

Step 3: Determine the Subnet Ranges

Since the subnet mask is **255.255.255.224**, the block size is

$$256 - 224 = 32$$

The subnet ranges are:

Subnet	Address Range
1	172.16.10.0 – 172.16.10.31
2	172.16.10.32 – 172.16.10.63
3	172.16.10.64 – 172.16.10.95
4	172.16.10.96 – 172.16.10.127
5	172.16.10.128 – 172.16.10.159
6	172.16.10.160 – 172.16.10.191
7	172.16.10.192 – 172.16.10.223
8	172.16.10.224 – 172.16.10.255

Step 4: Identify the Subnet of IP Address 172.16.10.78

The IP address **172.16.10.78** falls within the range

172.16.10.64 – 172.16.10.95

Therefore,

Network Address

172.16.10.64

Broadcast Address

172.16.10.95

Valid Host Range

172.16.10.65 – 172.16.10.94

Step 5: Verify IP Address 172.16.10.95

Given IP

172.16.10.95

This address is the **broadcast address** of the subnet **172.16.10.64/27**.

Therefore,

- It belongs to the **same subnet**.
- It **cannot be assigned to a host**, because it is reserved for broadcasting packets to all devices within the subnet.

6. Engineering Justification

Subnetting the **172.16.10.0/24** network into **/27** subnets creates **8 equal-sized subnets**, each supporting **30 usable hosts**. This approach efficiently divides the network into smaller broadcast domains, reducing unnecessary traffic and improving network performance. Identifying the correct subnet ensures accurate routing and prevents addressing conflicts.

7. Final Result

Parameter	Result
Original Network	172.16.10.0/24
Subnet Mask	255.255.255.224 (/27)
Total Number of Subnets	8
Total Addresses per Subnet	32
Usable Hosts per Subnet	30
IP Address	172.16.10.78
Subnet	172.16.10.64/27
Network Address	172.16.10.64
Valid Host Range	172.16.10.65 – 172.16.10.94
Broadcast Address	172.16.10.95
Does 172.16.10.95 belong to the same subnet?	Yes
Can 172.16.10.95 be assigned to a host?	No (Broadcast Address)

8. Calculation Summary

Parameter	Value
Borrowed Bits	3
Host Bits	5
Block Size	32
Total Subnets	8
Total Addresses/Subnet	32
Usable Hosts/Subnet	30

9. Engineering Interpretation

The **172.16.10.0/24** network has been successfully divided into **8 equal subnets** using the **/27** subnet mask. The IP address **172.16.10.78** belongs to the subnet **172.16.10.64/27**, whose valid host range is **172.16.10.65 to 172.16.10.94**. The address **172.16.10.95** is the broadcast address of the same subnet and therefore cannot be assigned to any host. This subnetting strategy improves network efficiency, minimizes broadcast traffic, and supports better network management.

5. A network uses the Internet Protocol version 6 address **2001:0db8:0000:0000:0000:ff00:0042:8329**. Apply IPv6 addressing rules to compress the given address into its shortest form and then expand the compressed address back to its full notation. Using a prefix length of **/64**, determine the network prefix of the given address. Also, calculate the total number of possible host addresses available within this network based on the given prefix length.

1. Problem Understanding

A network has been assigned the IPv6 address **2001:0db8:0000:0000:0000:ff00:0042:8329**. The task is to apply IPv6 addressing rules to compress the address into its shortest notation, expand it back into full notation, determine the network prefix using a **/64** prefix length, and calculate the total number of possible host addresses available in the network.

2. Given Data

Parameter	Value
IPv6 Address	2001:0db8:0000:0000:0000:ff00:0042:8329
Prefix Length	/64

3. Method Selection

The solution follows standard IPv6 addressing rules by:

- Removing leading zeros from hexadecimal blocks.
- Replacing one continuous sequence of zero blocks with " : :".
- Identifying the first **64 bits** as the network prefix.
- Calculating the available host addresses using the remaining **64 host bits**.

This method follows the IPv6 addressing standards defined for subnetting.

4. Formula Used

IPv6 Compression Rules

- Remove leading zeros from each hexadecimal block.
- Replace one continuous sequence of zero blocks with " : :".
- " : :" can be used only once in an IPv6 address.

Host Bits

Host Bits = 128 – Prefix Length

Total Host Addresses

Total Hosts = $2^{(\text{Host Bits})}$

5. Step-by-Step Solution

Step 1: Original IPv6 Address

2001:0db8:0000:0000:0000:ff00:0042:8329

Step 2: Compress the IPv6 Address

Remove Leading Zeros

2001:db8:0:0:0:ff00:42:8329

Replace Consecutive Zero Blocks

Three consecutive 0 blocks are replaced using " : :".

Compressed IPv6 Address

2001:db8::ff00:42:8329

Step 3: Expand the Compressed Address

Expanding " : :" restores the omitted zero blocks.

Expanded IPv6 Address

2001:0db8:0000:0000:0000:ff00:0042:8329

Step 4: Determine the Network Prefix

Prefix Length

/64

The first **64 bits** represent the network portion.

Network Prefix

2001:0db8:0000:0000::/64

Step 5: Calculate the Number of Host Addresses

Host Bits

128 – 64

= 64

Total Possible Host Addresses

2^{64}

= 18,446,744,073,709,551,616

Approximately

18.4 Quintillion Host Addresses

6. Engineering Justification

IPv6 uses **128-bit addresses**, providing an extremely large address space. A **/64 prefix** is the standard subnet size for IPv6 networks because it supports Stateless Address Auto Configuration (SLAAC), efficient routing, and compatibility with modern networking standards. This addressing scheme is suitable for large-scale enterprise and IoT deployments.

7. Final Result

Parameter	Result
Original IPv6 Address	2001:0db8:0000:0000:ff00:0042:8329
Compressed IPv6 Address	2001:db8:ff00:42:8329
Expanded IPv6 Address	2001:0db8:0000:0000:0000:ff00:0042:8329
Prefix Length	/64

Network Prefix	2001:0db8:0000:0000::/64
Host Bits	64
Total Possible Host Addresses	18,446,744,073,709,551,616 (2⁶⁴)

8. Calculation Summary

Parameter	Value
IPv6 Address Length	128 bits
Network Bits	64 bits
Host Bits	64 bits
Total Host Addresses	2 ⁶⁴
Decimal Value	18,446,744,073,709,551,616

9. Engineering Interpretation

The given IPv6 address has been successfully compressed to **2001:db8::ff00:42:8329** and expanded back to its original form. With a **/64 prefix**, the first 64 bits represent the network portion, while the remaining 64 bits are available for host addressing. This configuration supports **18,446,744,073,709,551,616** unique host addresses, making IPv6 highly scalable for modern networks, cloud computing, IoT devices, and future Internet expansion.

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Problem Understanding	20%	Clearly interprets problem, identifies all parameters and requirements accurately	Understands problem with minor gaps	Partial understanding; misses key elements	Misinterprets or unable to understand problem
Method Selection	20%	Selects most appropriate method/formula with strong justification	Selects correct method with minor errors	Inappropriate or partially correct method	Unable to select method
Solution Process	25%	Logical, systematic steps with correct approach throughout	Mostly correct steps with minor mistakes	Disorganized steps; multiple errors	No clear process
Accuracy of Calculation	20%	All calculations accurate; correct final answer	Minor calculation errors	Frequent errors; incorrect final answer	No valid calculations
Interpretation of Results	15%	Clearly interprets results with units and engineering relevance	Basic interpretation with minor omissions	Limited or unclear interpretation	No interpretation